

CHICAGO EXECUTIVE, IL, USA

WMO: 744665

Lat: 42.121N

Lon: 87.905W

Elev: 636

StdP: 14.36

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04838

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.0	3.5	-11.4	3.0	0.5	-7.2	3.8	5.3	24.3	30.1	21.5	30.0	8.8	270	0.521

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.4	91.4	74.0	88.9	72.7	86.2	71.5	76.8	87.2	75.1	84.8	73.6	82.7	11.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.3	126.9	82.8	72.1	121.4	81.2	70.4	114.7	79.5	40.5	87.6	38.9	84.7	37.5	82.9	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	21.4	19.0	17.2	DB	-7.4	95.9	6.9	3.3	-12.4	98.3	-16.4	100.2	-20.3	102.1	-25.3	104.5
(5)				WB	-8.1	79.2	6.4	1.5	-12.7	80.3	-16.4	81.2	-20.0	82.0	-24.6	83.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	50.5	24.4	26.6	38.0	48.8	58.7	69.3	74.4	72.8	66.1	53.4	41.2	30.4	(6)
(7)		DBStd	19.65	11.81	10.89	10.97	9.28	9.25	7.42	6.08	5.40	7.65	8.72	9.37	10.25	(7)
(8)		HDD50	2969	795	658	403	131	20	0	0	0	0	63	288	611	(8)
(9)		HDD65	6203	1260	1076	841	495	245	39	3	6	78	374	713	1073	(9)
(10)		CDD50	3139	1	2	30	94	290	579	757	707	483	168	26	2	(10)
(11)		CDD65	896	0	0	3	8	50	168	295	248	110	14	0	0	(11)
(12)		CDH74	8054	0	0	25	108	532	1637	2787	1934	915	116	0	0	(12)
(13)		CDH80	2739	0	0	4	21	158	586	1040	616	291	23	0	0	(13)
(14)	Wind	WSAvg	7.8	9.1	8.8	8.6	9.2	7.8	6.9	6.4	5.9	6.4	7.7	8.7	8.6	(14)
(15)	Precipitation	PrecAvg	36.7	1.9	1.5	2.7	3.9	3.5	4.0	3.9	3.6	3.9	2.6	2.8	2.5	(15)
(16)		PrecMax	47.5	5.2	3.5	5.3	6.3	8.3	8.0	11.1	7.8	13.9	8.0	7.1	5.6	(16)
(17)		PrecMin	26.7	0.3	0.2	0.4	1.2	1.4	0.7	1.3	0.8	0.0	0.2	0.5	0.4	(17)
(18)		PrecStd	5.4	1.2	0.9	1.2	1.5	1.6	1.7	2.0	2.3	3.0	1.7	1.5	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.2	60.9	77.6	82.9	88.9	93.4	96.7	93.2	91.2	83.2	68.8	60.9	(19)
(20)		MCWB	53.2	52.3	63.0	64.5	69.5	74.0	75.9	74.1	73.6	69.4	57.8	56.2	(20)	
(21)		2%	DB	49.9	53.4	67.8	76.8	84.5	90.3	91.9	89.7	87.4	77.0	63.7	53.8	(21)
(22)		MCWB	44.5	46.6	57.2	60.9	67.6	72.6	74.8	73.5	71.4	64.8	56.0	49.7	(22)	
(23)		5%	DB	43.5	45.7	61.1	71.6	80.3	86.6	88.8	86.3	83.0	72.5	60.4	48.2	(23)
(24)		MCWB	39.1	40.4	52.9	57.8	65.6	71.3	73.6	72.0	69.0	62.7	53.5	44.3	(24)	
(25)		10%	DB	39.3	40.9	54.7	65.9	75.4	82.8	85.8	83.1	79.5	67.8	55.7	44.8	(25)
(26)		MCWB	36.5	36.9	47.3	54.5	64.0	69.4	72.1	70.4	67.3	59.7	49.6	41.0	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.8	54.6	64.2	66.6	73.2	77.7	79.4	77.4	75.6	71.5	61.9	56.8	(27)
(28)		MCDWB	56.6	59.0	76.2	78.8	85.0	89.6	91.6	88.9	86.8	80.9	65.3	60.1	(28)	
(29)		2%	WB	45.7	46.9	59.1	62.9	70.2	74.9	77.3	75.6	73.5	67.8	58.0	50.0	(29)
(30)		MCDWB	48.7	52.5	66.0	74.0	80.4	85.2	87.5	84.5	82.6	73.2	62.7	52.8	(30)	
(31)		5%	WB	40.0	41.3	53.9	59.6	67.9	73.0	75.7	74.2	71.4	64.2	53.8	45.0	(31)
(32)		MCDWB	42.9	45.9	60.6	69.6	77.2	83.6	85.8	82.4	80.5	70.6	58.4	47.5	(32)	
(33)		10%	WB	36.7	37.0	47.4	55.5	65.2	71.1	74.0	72.5	69.3	60.6	50.2	40.9	(33)
(34)		MCDWB	39.7	40.5	54.8	66.0	73.8	80.8	83.1	80.4	77.2	66.7	55.7	44.1	(34)	
(35)	Mean Daily Temperature Range	MDBR	13.4	14.4	16.0	18.4	19.1	18.8	17.4	16.7	17.9	16.6	14.7	12.5	(35)	
(36)		5% DB	MCDBR	17.5	20.1	24.3	28.3	25.4	22.3	20.0	20.5	22.0	22.9	19.9	17.0	(36)
(37)		MCWBR	14.8	15.3	16.6	16.6	13.1	10.4	8.5	8.7	9.9	13.1	14.5	13.9	(37)	
(38)		5% WB	MCDBR	16.6	19.6	22.8	25.3	22.4	20.3	18.0	17.4	19.1	18.9	17.6	15.5	(38)
(39)		MCWBR	14.7	15.6	16.8	16.9	12.9	11.0	9.2	8.8	9.6	13.5	14.8	14.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.297	0.312	0.344	0.399	0.426	0.439	0.443	0.437	0.408	0.355	0.314	0.296	(40)	
(41)		taud	2.441	2.392	2.390	2.260	2.240	2.252	2.240	2.255	2.305	2.454	2.515	2.477	(41)	
(42)		Ebn at Noon	271	284	286	276	270	266	263	261	260	264	262	261	(42)	
(43)		Edh at Noon	24	30	34	41	43	43	43	41	36	28	23	22	(43)	
(44)	All-Sky Solar Radiation	RadAvg	491	771	1131	1464	1676	1919	1937	1685	1385	904	571	409	(44)	
(45)		RadStd	59	96	91	120	134	128	94	101	118	101	57	45	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page